A/B Compartment Analyses

1. Load Hi-C Files

Similar to the TAD analyses, we used the cooler tool (version 0.10.4) to load the Hi-C files to prepare them for downstream analyses. The `cooler merge` commands merge two 80 kb resolution Hi-C contact matrices for the infected (`merged_cov_HiC.mcool`) and mock-treated (`merged_mock_HiC.mcool`) conditions into separate merged `.mcool` files. The `cooler balance` commands then perform matrix balancing on each merged dataset. We restricted our analysis to chromosome 9 at 80kb resolution to reproduce Fig 1c in particular, and it also helps in reducing computational complexity.

```
cooler merge merged_cov_HiC.mcool \
  GSM5411175_NS74-NS67-HiC7.mcool::/resolutions/80000 \
  GSM5411176_NS74-NS67-HiC8.mcool::/resolutions/80000

cooler merge merged_mock_HiC.mcool \
  GSM5411173_NS74-NS67-HiC1.mcool::/resolutions/80000 \
  GSM5411174_NS74-NS67-HiC2.mcool::/resolutions/80000

cooler balance merged_mock_HiC.mcool
cooler balance merged_cov_HiC.mcool
```

2. GC track for eigenvector phasing

As described in the paper, A/B nuclear compartments were identified on the basis of decomposed eigenvectors (E1) from Hi-C contact matrices using cooltools. A/B compartmental scores (E1) were corrected by GC densities in each bin.

To do so, we used cooltools (version 0.7.1, the paper used version 0.4.1 but it is unsupported on our system, and the results are reproduced) and downloaded the hg38 reference. The paper used the hg19 reference, but we managed to reproduced the result with a different reference.

We generated the genomic annotation files at 80 kb resolution on chromosome 9. We calculated chromosome sizes for the hg38 reference genome and cleaned the file to retain only standard chromosomes with numeric lengths. The genome is then partitioned into 80 kb bins using `cooltools genome binnify`, and filtered for bins corresponding to chromosome 9. Because different tools use different chromosome naming conventions, we temporarily renamed `chr9` to the NCBI accession (`CM000671.2`) so that GC content can be calculated from the corresponding NCBI reference genome FASTA. After GC content is computed for each genomic bin, we converted the chromosome name back to the standard `chr9` format. We created a chromosome view file defining the genomic coordinates of chromosome 9 to provide a reference region for subsequent analyses (compartment calling or visualization).

```
cooltools genome fetch-chromsizes hg38 > hg38.chrom.sizes

# cleaning
awk 'BEGIN{OFS="\t"} $2 ~ /^[0-9]+$/ {print $1, $2}' \
  hg38.chrom.sizes > hg38.clean.chrom.sizes
cooltools genome binnify hg38.clean.chrom.sizes 80000 > hg38.80000.bins.bed

# filter for chromosome 9
awk '$1=="chr9"' hg38.80000.bins.bed > chr9.80000.bins.bed

# standardizing naming conventions
awk 'BEGIN{OFS="\t"} NR==1{$1="chrom"; print; next} \
  $1=="chr9"{$1="CM000671.2"; print}' \
  chr9.80000.bins.bed > chr9.80000.bins.ncbi.bed

cooltools genome gc chr9.80000.bins.ncbi.bed \
  ncbi_dataset/ncbi_dataset/data/GCA_000001405.15/GCA_000001405.15_GRCh38_genomic.fna \
  > chr9.80000.gc.tsv

# standardizing naming conventions
awk 'BEGIN{OFS="\t"} NR==1{print; next} {$1="chr9"; print}' \
  chr9.80000.gc.tsv > chr9.80000.gc.chrnames.tsv

echo -e "chr9\t0\t138394717\tchr9" > chr9.view.tsv
```

3. Compartment analysis

Saddle plot analyses were performed to measure the compartmentalization strength on a genome-wide scale using cooltools. We first used `cooltools eigs-cis` to perform eigenvalue decomposition to calculate compartment signal by finding the eigenvector that correlates best with the phasing track, using GC content to orient the compartments and producing compartment tracks such as E1. Next, we used `cooltools expected-cis` to calculate the expected Hi-C signal for cis regions of chromosomal interaction maps, and then `cooltools saddle` using the mock E1 compartment ordering to generate saddle plots for both mock and CoV samples. This allows for comparison of A/B compartment strength and organization under the same reference ordering. Then, we used `cooler dump`, `cooler load`, and `cooler balance` to extract only chr9 bins and contacts from the mock dataset, rebuild them as a standalone `.cool` file, and normalize it.

```
for SAMPLE in mock cov; do
  cooltools eigs-cis \
    merged_${SAMPLE}_HiC.mcool \
    --view chr9.view.tsv \
    --phasing-track chr9.80000.gc.chrnames.tsv::GC \
    --n-eigs 3 \
    -o ${SAMPLE}_80000
done


for SAMPLE in mock cov; do
  cooltools expected-cis \
    --view chr9.view.tsv \
    merged_${SAMPLE}_HiC.mcool \
    -p 8 \
    -o ${SAMPLE}_chr9_80000.expected.tsv

  cooltools saddle \
    --view chr9.view.tsv \
    merged_${SAMPLE}_HiC.mcool \
    mock_80000.cis.vecs.tsv::E1 \
    ${SAMPLE}_chr9_80000.expected.tsv::balanced.avg \
    -n 50 \
    --qrange 0.02 0.98 \
    --fig png \
    --strength \
    -o ${SAMPLE}_chr9_80000.mockOrdered.saddle
done

cooler dump -t bins merged_mock_HiC.mcool | awk '$1=="chr9"' > chr9.bins.tsv
cooler dump -t pixels merged_mock_HiC.mcool --join \
  | awk '$1=="chr9" && $4=="chr9"{print $2"\t"$5"\t"$7}' > chr9.pixels.tsv

cooler load -f coo chr9.bins.tsv chr9.pixels.tsv merged_mock_chr9.cool
```

```
cooler balance merged_mock_chr9.cool
```

4. Correlation maps from observed/expected Hi-C contact matrices

We generated Pearson correlation maps from observed/expected Hi-C contact matrices for chromosome 9 in the mock and CoV conditions. We computed the expected cis contact frequencies for chr9, extracted the chr9 region from 100–130 Mb (for Fig 1c), and converted the matrix into an observed/expected matrix by normalizing each diagonal by its expected value. We then plotted the resulting correlation matrix.

```python
# pearson_compartments.py

import cooler
import cooltools
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

def corr_pairwise_nan(oe):
    n = oe.shape[0]
    corr = np.full((n, n), np.nan)

    for i in range(n):
        xi = oe[:, i]
        for j in range(i, n):
            xj = oe[:, j]
            valid = np.isfinite(xi) & np.isfinite(xj)

            if valid.sum() >= 3:
                corr_ij = np.corrcoef(xi[valid], xj[valid])[0, 1]
                corr[i, j] = corr_ij
                corr[j, i] = corr_ij

    return corr

def pearson_oe(clr_path, chrom, out_png):
    clr = cooler.Cooler(clr_path)

    view_df = pd.DataFrame({
        "chrom": [chrom],
        "start": [0],
```

```
    "end": [clr.chromsizes[chrom]],
    "name": [chrom],
})

exp = cooltools.expected_cis(
    clr,
    view_df=view_df,
    nproc=4,
)

res = clr.binsize

start_bp = 100_000_000
end_bp = 130_000_000

mat = clr.matrix(balance=True).fetch(
    (chrom, start_bp, end_bp)
).astype(float)

chrom_exp = exp[exp["region1"] == chrom].set_index("dist")["balanced.avg"]

n = mat.shape[0]

oe = np.full(mat.shape, np.nan)

for d in range(n):
    val = chrom_exp.get(d, np.nan)
    if np.isfinite(val) and val > 0:
        i = np.arange(n - d)
        oe[i, i + d] = mat[i, i + d] / val
        oe[i + d, i] = mat[i + d, i] / val

corr = corr_pairwise_nan(oe)

plt.figure(figsize=(7, 6))
plt.imshow(
    corr,
    cmap="RdBu_r",
    origin="upper",
    extent=(100, 130, 130, 100),
)
plt.xlabel("Position (Mb)")
```

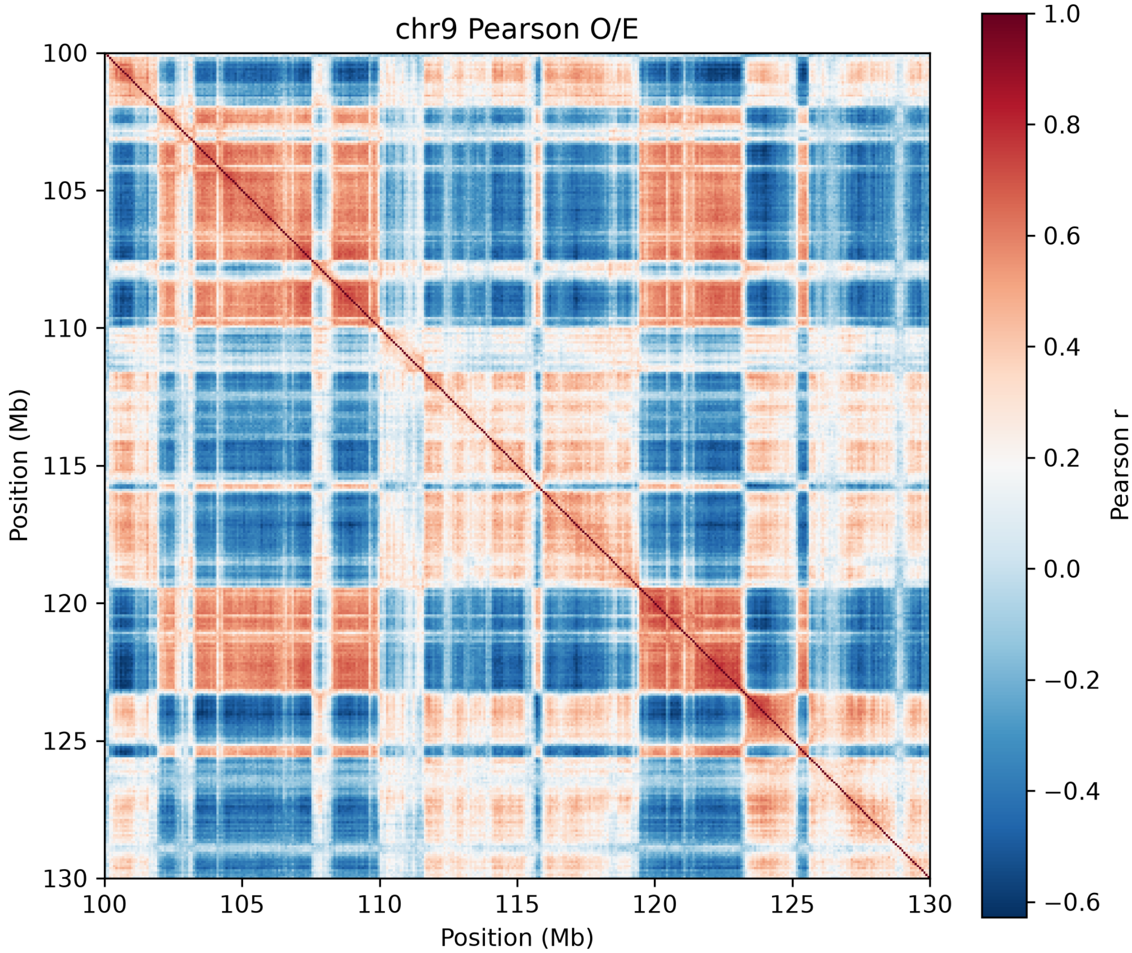
```
    plt.ylabel("Position (Mb)")
    plt.colorbar(label="Pearson r")
    plt.title(f"{chrom} Pearson O/E")
    plt.tight_layout()
    plt.savefig(out_png, dpi=300)
    plt.close()

pearson_oe("merged_mock_HiC.mcool", "chr9", "mock_chr9_pearson.png")
pearson_oe("merged_cov_HiC.mcool", "chr9", "cov_chr9_pearson.png")
```

With reference to the figures below, we reproduced Fig 1c. Shown below are the Pearson correlation matrices of Hi-C of a region (chr9: 100–130 Mb, hg19), where they show alteration of the 3D genome and chromatin compartmentalization by SARS-COV-2. These matrices are used to visualize chromatin compartmentalization (A and B compartments) by measuring how similarly different genomic regions interact with the rest of the chromosome. From the matrices and the right panel of Fig 1c, we see that there is increased A-B mixing and reduced A-A interactions, indicating that interactions between active (A) and inactive (B) chromatin compartments become more similar after infection and there is a weakening of interactions among transcriptionally active chromatin regions. These observations demonstrate that SARS-CoV-2 disrupts large-scale chromatin organization by weakening compartment structure. Instead of maintaining distinct active and inactive chromatin domains, infected cells exhibit increased mixing between compartments and reduced cohesion within active chromatin

Our results differ from the paper's in one area: we obtained negative correlations in regions where the paper obtained zero correlations. We propose that this reflects differences in processing that the paper did not mention in detail. In our code, Pearson correlations are computed directly from the observed/expected matrix, so bins with opposite interaction patterns can produce negative values. The result from the paper may mask negative correlations or used a different normalization. We could also have used different balancing weights, expected-cis calculation, or differed in whether low-coverage bins were filtered. It would help if the authors could describe their method in greater detail to explain this discrepancy.

Pearson correlation matrix of Hi-C: Mock



Pearson correlation matrix for Hi-C: SARS-COV-2